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MOTOR

Abstract

The present disclosure relates to a motor. The motor is equipped with a shaft center cooling structure. The motor includes a rotor shaft that includes, on a shaft center, at least two kinds of a first coolant channel and a second coolant channel. The coolant channels are each configured separately along a rotation axis direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
[0001] This application claims priority to Japanese Patent Application No. 2024-033330 filed on Mar. 5, 2024, incorporated herein by reference in its entirety.
BACKGROUND
1. Technical Field
[0002] The present disclosure relates to a motor.
2. Description of Related Art
[0003] Japanese Unexamined Patent Application Publication No. 2022-30847 (JP 2022-30847 A) describes technology of a rotating electric machine including a rotor having a hollow rotor shaft and a rotor core, a stator having a stator core and a coil, and a case member housing the rotor and the stator, in which a coolant path for cooling outside of the stator with coolant is formed. In this technology, an oil channel structure for supplying oil, such as automatic transmission fluid (ATF) or the like, discharged by an oil pump, is provided inside the hollow of the rotor shaft, and heat-exchange is performed between the coolant flowing in the coolant path of the case member, and the oil discharged by the oil pump. This ensures cooling performance for the rotating electric machine.
SUMMARY
[0004] However, in JP 2022-30847 A, the inside of the rotor is cooled by an oil path of one type of oil, and thus the cooling performance regarding the rotor is becomes lower when the oil is at high temperatures, and also heat exchange between the oil and the coolant is performed at another place outside the rotor, and accordingly there is room for improvement.
[0005] The present disclosure provides a motor capable of maintaining cooling performance.
[0006] A motor according to the present disclosure is equipped with a shaft center cooling structure, the motor including a rotor shaft that includes, on a shaft center, at least two kinds of coolant channels that are each configured separately along a rotation axis direction.
[0007] The present disclosure provides an effect that cooling performance can be improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:
[0009] FIG. **1** is a cross-sectional view showing a schematic configuration of a motor according to an embodiment; and
[0010] FIG. **2** is a cross-sectional arrow view of the rotor and rotor shaft taken along A-A of FIG. **1**.
DETAILED DESCRIPTION OF EMBODIMENTS
[0011] Hereinafter, a motor according to an embodiment of the present disclosure will be described with reference to the drawings. The constituent elements in the following embodiments include those that can be easily replaced by a person skilled in the art or those that are substantially the same. In addition, the drawings referred to in the following description only schematically show shapes, sizes, and positional relationships to the extent that the contents of the present disclosure can be understood. That is, the present disclosure is not limited to only the shapes, sizes, and positional relationships illustrated in the drawings.
Motor Configuration
[0012] FIG. **1** is a cross-sectional view illustrating a schematic configuration of a motor according to an embodiment. The motor **1** shown in FIG. **1** is mounted on a vehicle, for example, and is rotationally driven by a current supplied from the outside, for example, a three-phase alternating current. The motor **1** includes a substantially cylindrical stator **2** fixed to a frame (housing) or the

like (not shown), a rotor **3** rotatably held on an inner peripheral side of the stator **2**, and a rotor shaft **4**.

[0013] FIG. **2** is a cross-sectional arrow view of the rotor **3** and the rotor shaft **4** taken along A-A of FIG. **1**. FIG. **1** is a cross-sectional view taken along B-B line of FIG. **2**. In FIG. **1** and FIG. **2**, the axial direction of the rotor shaft **4** is defined as the X direction, the circumferential direction orthogonal to the rotor shaft **4** is defined as the Y direction, and the depth direction orthogonal to the rotor shaft **4** is defined as the Z direction.

[0014] As shown in FIGS. **1** and **2**, the stator **2** includes a stator core **21** and a stator coil **22**. The stator **2** is fixed to an inner peripheral surface side of a frame (housing) (not shown).

[0015] The stator core **21** is configured by using an annular member in which a plurality of electromagnetic steel plates as a plate material made of a magnetic material are laminated. The stator core **21** is formed by setting an inner diameter so as to have an annular gap (air gap) between its inner peripheral surface and the rotor **3**.

[0016] The stator coil **22** is inserted into a slot (not shown) formed in the stator core **21**, and forms a magnetic pole corresponding to a current supplied from the outside. The stator coil **22** is composed of a first stator coil to which a U-phase current is supplied, a second stator coil to which a V-phase current is supplied, and a third stator coil to which a W-phase current is supplied, among three-phase alternating currents. The third stator coil is sequentially arranged in the circumferential direction of the stator core **21** from the first stator coil.

[0017] As shown in FIGS. **1** and **2**, the rotor **3** is attached to the rotor shaft **4** and is configured to be rotatable about a rotation axis Xa of the rotor shaft **4**. The rotor **3** includes a rotor core **31**, a rotor coil **32**, and a plurality of coolant channels **33**.

[0018] Similarly to the stator core **21** described above, the rotor core **31** is configured by using an annular member in which a plurality of electromagnetic steel plates as a plate material made of a magnetic material are laminated. The rotor core **31** is formed by setting an outer diameter so as to have the above-described gap (air gap).

[0019] The rotor coil **32** is formed and accommodated in a slot (not shown) of the rotor core **31** by winding a coil element wire around each of a plurality of teeth (not shown) formed in the rotor core **31**. In FIG. **1** and FIG. **2**, only symbols are used.

[0020] The plurality of coolant channels **33** are formed in the rotor core **31** along the rotation axis Xa of the rotor shaft **4**, and discharge oil OL as coolant from both ends of the rotor core **31**. Here, as the oil OL, for example, ATF or the like is used. Further, the plurality of coolant channels **33** are provided at four positions at predetermined intervals, for example, at intervals of 90 degrees along the circumferential direction around the rotation axis Xa of the rotor shaft **4**. Further, the plurality of coolant channels **33** are partially branched so as to be connectable to a coolant channel provided in the rotor shaft **4**, which will be described later, and oil OL from the coolant channel provided in the rotor shaft **4** flows in.

[0021] As shown in FIGS. **1** and **2**, the rotor shaft **4** has a fixed shaft **41** and a fixed shaft **41** in which one axial end is fixed to a case (housing) (not shown), and a rotating shaft **42** which rotates about a rotation axis Xa.

[0022] The fixed shaft **41** has a cylindrical shape (hollow portion), and has a channel **411** through which coolant Wa supplied from an external pump (not shown) flows. The fixed shaft **41** has an outer diameter smaller than an inner diameter of the rotating shaft **42** so as to have a gap (air gap).

[0023] The rotating shaft **42** is an annular rod-shaped member having an inner diameter larger than an outer diameter of the fixed shaft **41**, and is rotatably supported by a support point **43** such as a bearing provided in a case (housing) (not shown). In the rotating shaft **42**, the rotor **3** is attached to the outer peripheral side, the drive transmission component **5** is attached to the inner peripheral portion of the right end portion, and the rotational driving force of the rotor **3** is transmitted to the drive transmission component **5**.

[0024] Further, the rotating shaft **42** is formed so that the axial lengths of the hole **421** on the inner peripheral side and the fixed shaft **41** differ so that a gap K**1** is formed between the hole **421** on the inner peripheral side and the distal end of the fixed shaft **41**. As a result, in the rotating shaft **42**, the coolant Wa from the channel **411** of the fixed shaft **41** is discharged to the gap K**1** between the fixed shaft **41** and the hole **421**, and the coolant Wa flows into one axial end (left end) of the rotor shaft **4** via the hole **421**. That is, the channel **411** and the channel **422** form the first coolant channel **6** that cools the motor **1** by circulating the coolant Wa supplied from the cooling pump (not shown) into the cooling pump. As the coolant Wa, for example, LLC (Long Life Coolant) or the like is used.

[0025] Further, the rotating shaft **42** has a plurality of channels **423** that are outward of the first coolant channel **6** in the rotation axis Xa and through which oil OL functioning as a coolant for cooling the motor **1** along the circumferential direction around the rotation axis Xa flows. Each of the plurality of channels **423** is provided at four positions of the rotating shaft **42** at a predetermined interval, for example, an interval of 90 degrees along the circumferential direction around the rotation axis Xa of the rotor shaft **4**. Further, each of the plurality of channels **423** has a section D**1** that overlaps with a portion extending along the rotation axis Xa. Specifically, each of the plurality of channels **423** partially overlaps with each other so as to surround a section D**1** of a part of the first coolant channel **6** from the outer periphery along the rotation axis Xa. More specifically, each of the plurality of channels **423** is provided in the rotating shaft **42** so as to overlap with a section D**1** of a part of the first coolant channel **6** along the rotation axis Xa. Further, each of the plurality of channels **423** is connected to each of the plurality of coolant channels **33** provided in the rotor core **31**, and supplies oil OL from an oil pump or a drive gear (not shown) to each of the plurality of coolant channels **33**. That is, the plurality of channels **423** and the plurality of coolant channels **33** provided in the rotor core **31** circulate the oil OL to form the second coolant channel **7** for cooling the motor **1**.

[0026] As described above, the second coolant channel **7** and the first coolant channel **6** are formed separately along the rotation axis Xa in the rotor shaft **4**. Therefore, the cooling effect can be maintained even when either the coolant Wa or the oil OL is heated. Further, the second coolant channel **7** and the first coolant channel **6** can heat-exchange the coolant Wa with the oil OL in a partial overlapping section D**1** extending along the rotation axis Xa. Consequently, the motor **1** can maintain the cooling effectiveness even when either the coolant Wa or the oil OL is hot.

[0027] According to the embodiment described above, the first coolant channel **6** and the second coolant channel **7** are formed separately along the rotation axis Xa in the rotor shaft **4**. Therefore, the cooling effect can be maintained even when either the coolant Wa or the oil OL is heated.

[0028] In addition, according to the embodiment, the coolant Wa and the oil OL can be heat-exchanged in the section D**1** in which the second coolant channel **7** and the first coolant channel **6** partially overlap each other extending along the rotation axis Xa. Therefore, the cooling effect can be maintained even when either the coolant Wa or the oil OL is heated.

[0029] In the embodiment, the coolant Wa is circulated in the first coolant channel **6**, but the type of the coolant Wa circulating in the first coolant channel **6** can be changed as appropriate, and an oil OL may be used as long as the channel is different.

[0030] In addition, in one embodiment, the rotor shaft **4** is formed separately of only the first coolant channel **6** and the second coolant channel **7**, but the present disclosure is not limited thereto, and a coolant channel in which another type of coolant circulates may be further provided in the rotor shaft **4**.

[0031] Additional benefits and variations can be readily derived by one of ordinary skill in the art. The broader aspects of the disclosure are not limited to the specific details and representative embodiments presented and described above. Accordingly, various modifications may be made without departing from the spirit or scope of the general inventive concept as defined by the appended claims and their equivalents.

[0032] While some of the embodiments of the present application have been described in detail with reference to the drawings, these are merely examples, and the present disclosure can be implemented in other forms in which various modifications and improvements are made based on the knowledge of a person skilled in the art, including the aspects described in the section of the disclosure of the present disclosure.

Claims

1. A motor equipped with a shaft center cooling structure, the motor comprising a rotor shaft that includes, on a shaft center, at least two kinds of coolant channels that are each configured separately along a rotation axis direction.

2. The motor according to claim 1, wherein: in the rotor shaft, the at least two kinds of coolant channels include a first coolant channel and a second coolant channel; and each of the first coolant channel and the second coolant channel includes a section that overlaps at a portion extending along the rotation axis direction.